Local root conditioning and companion eigenvectors of Gaussian polynomials

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Abstract

For a polynomial of degree n with independent standard complex Gaussian coefficients, we prove a degree-independent second-power tail for the eigenvector condition number of its companion matrix at a uniformly selected root. Its moments are uniformly bounded precisely below order two, while its second moment is $2\log n + O(1)$. The expected logarithm of the relative companion-to-coefficient condition ratio is $\log n + O(1)$. The proof separates local root sensitivity from interpolation in the companion basis. For Gaussian systems, marked Kac–Rice and a kernel metric reduce the normalized Jacobian to an induced Ginibre law, giving a sharp fourth-power conditioning tail. Exact box root laws and a tridiagonal Gram matrix then yield the companion results. We specify the perturbation spaces and output metrics throughout; the comparison concerns local conditioning, rather than stability of a complete algorithm.

1 Introduction

Computing the roots of a polynomial by solving a companion eigenvalue problem changes the perturbations to which the answer is exposed. A polynomial root responds to errors in its coefficients; the unstructured eigenvalue condition permits errors in every entry of the companion matrix. We ask how the conditioning of these two problems compares for Gaussian input. Our main result gives a sharp probabilistic description of companion eigenvalue conditioning and a relative comparison with the original root problem.

Average-condition theorems for matrix polynomials use their coefficient matrices as input. Applying such a theorem after linearization also requires its probability law and perturbation norm to match those of the induced companion problem. Section 1.2 makes this distinction explicit for the results of Armentano–Beltrán.

Consider

$$p(x) = \sum_{k=0}^{n} a_k x^k, \qquad a_0, \dots, a_n \text{ independent standard complex Gaussians.}$$

Here each coefficient has independent real and imaginary parts of variance $1/2$. Let C_p be the $n \times n$ monic companion matrix, with ones on its subdiagonal and last column $-(a_0, \dots, a_{n-1})^T/a_n$. First sample p, then choose one of its n roots Z uniformly; write $\mathbb{P}_*, \mathbb{E}_*$ for this joint law. Almost surely the roots are simple and nonzero.

For a root z, let r, w be right and left eigenvectors of C_p at z: $C_p r = zr$ and $w^* C_p = z w^*$. The quantity

$$\Lambda(p, z) = \frac{\|w\|_2 \|r\|_2}{|w^* r|} \tag{1}$$

is the absolute eigenvalue condition number for arbitrary matrix perturbations in spectral norm. It is often called the eigenvector overlap. We compare relative condition numbers using Euclidean norm on polynomial coefficients, spectral norm on matrices, and relative error in the same scalar root. Their precise definitions are given in Section 2.

Theorem 1.1 (Main result: conditioning of Gaussian companions)**.** *There is an absolute constant C such that the following hold.*

(i) For every degree $n \geq 1$ and every $t > 0$,

$$\mathbb{P}_*\{\Lambda(p, Z) > t\} \leq \min\{1, Ct^{-2}\}. \tag{2}$$

Moreover,

$$\mathbb{E}_*\Lambda(p, Z)^2 = 2\log n + O(1) \qquad (n \to \infty). \tag{3}$$

Consequently the moments $\mathbb{E}_*\Lambda^q$ are bounded uniformly in degree precisely for $0 < q < 2$, and $\mathbb{E}_* \log \Lambda = O(1)$.

(ii) The relative eigenvalue and coefficient condition numbers satisfy

$$\mathbb{E}_* \log \frac{\kappa_{\mathrm{eig,rel}}(C_p, Z)}{\kappa_{\mathrm{coeff,rel}}(p, Z)} = \log n + O(1) \qquad (n \to \infty). \tag{4}$$

The constants in the $O(1)$ terms are independent of n.

The first assertion controls the expected fraction of roots with a large eigenvalue condition number, uniformly in the degree. Its exponent is sharp: any larger exponent with a degree-independent constant would bound the second moment, contradicting (3). The second assertion says that the geometric mean of the relative companion-to-coefficient condition ratio is comparable to n. These statements fit together because the coefficient root condition has its own degree scale. They concern a uniformly selected root and two specified local problems, rather than a bound on all errors in a complete rootfinding algorithm.

For clarity, the two relative conditions in the theorem are

$$\kappa_{\mathrm{coeff,rel}}(p, z) = \frac{\|a\|_2 \big(\sum_{k=0}^{n} |z|^{2k}\big)^{1/2}}{|z|\,|p'(z)|}, \qquad \kappa_{\mathrm{eig,rel}}(C_p, z) = \Lambda(p, z)\frac{\|C_p\|_2}{|z|}, \tag{5}$$

where $a = (a_0, \ldots, a_n)^T$ and every polynomial coefficient is perturbable. Theorem 5.1 proves part (i), and Corollary 5.7 proves part (ii).

1.1 Proof strategy and the role of root geometry

The proof separates local root sensitivity from the additional conditioning of interpolation in the companion basis. To analyze local sensitivity, we first work with systems of d independent complex Gaussian polynomials sharing a covariance kernel. That kernel defines a natural output metric. Marked Kac–Rice shows that, after normalization by this metric, the Jacobian at a selected root is independent of the root location and has a determinant-weighted Gaussian law. The resulting intrinsic condition number κ_H, defined in Section 2.3, satisfies

$$\mathbb{P}_*\{\kappa_H > t\} \leq \min\left\{1, \frac{d(d+1)}{4t^4}\right\},$$

with the constant asymptotically exact for fixed d. The precise assumptions and statements are Theorems 3.3 and 3.4. The induced Ginibre and Wishart distributions used here are established random-matrix results [15, 17].

For the Gaussian box ensemble, exact radial laws then translate this geometric normalization back to Euclidean root displacement. They give $\mathbb{E}_* \log \kappa_{\mathrm{coeff}} = -\log n + O_d(1)$ and retain the contribution of roots outside the unit polydisk. Finally, in one variable, writing $p(x) = (x - z)q(x)$ turns the coefficient norm into the Gram matrix of multiplication by $x - z$. This matrix is tridiagonal. Its inverse trace, together with the Gaussian weight at a root, gives the exact conditional overlap moment and the bounds in Theorem 1.1. The same argument also shows that, for each fixed $n \geq 2$, overlap moments are finite precisely below order four; the threshold two concerns uniformity as the degree grows.

The paper follows this proof structure. Section 2 fixes the input–output maps, metrics, and sampling law. Section 3 establishes the local Gaussian law; Section 4 derives the box root and coefficient scales; and Section 5 proves the main theorem. Section 6 discusses its interpretation and the multivariate questions that remain.

1.2 Relation to previous work

Local conditioning of polynomial systems has a substantial geometric and probabilistic theory. Shub–Smale developed the projective framework, its relation to ill-posedness, and distributional estimates in the *Complexity of Bézout's theorem* series [23, 24, 25]. Bürgisser–Cucker [11] give a systematic account. Proposition 2.2 is a standard derivative-norm calculation in explicitly specified input and output metrics; it is included to fix conventions, not as a new condition-number theorem.

Beltrán–Pardo [6, 7] used average conditioning and integration over system–root pairs to obtain average polynomial-time homotopy algorithms. Their work also gives exact moments of the normalized projective condition number below order four [7, Sections 7–8]. Bürgisser–Cucker [10] established smoothed estimates, and Armentano–Beltrán–Bürgisser–Cucker–Shub [2, Theorem 2] obtained an exact centered inverse-Jacobian Frobenius second moment and a smoothed bound. Section 3.4 spells out the overlap with our local law. Lairez [18] subsequently obtained a deterministic algorithm with polynomial average complexity.

For sparse systems, Malajovich–Rojas [21] construct metrics from the coefficient spaces, relate conditioning to the discriminant, and prove fourth-power tail bounds. Their normalization and worst-root probability differ from our absolute, selected-root formulation. Toric condition metrics and homotopy complexity are developed further in [19, 20]. Our local section presents the common-kernel Gaussian calculation in a form suited to comparing coefficient and algebraic-representation perturbations; the fourth-power exponent and the homogeneous moment identities already belong to this established theory.

Toh–Trefethen [26] explain the different perturbation geometries of polynomial roots and companion eigenvalues. Noferini–Townsend and Graf–Townsend construct multivariate examples in which intermediate algebraic problems amplify conditioning exponentially with the number of variables [22, 16]. Our representation theorem concerns one variable and growing degree; the local framework extends to systems, but the present paper does not resolve those growing-dimension examples.

Bürgisser–Cucker–Rocha Cardozo [12] study the conditioning of characteristic polynomials induced by Gaussian matrices. Their results demonstrate the importance of the induced coefficient law in an algebraic reduction. Here the construction runs from a Gaussian polynomial to its companion matrix, giving a different probability distribution and perturbation problem.

Armentano–Beltrán [1, Theorem 1.1] study the polynomial eigenvalue problem for $Q(z) = \sum_{j=0}^{k} B_j z^j$, where $B_j \in \mathbb{C}^{m\times m}$. Their perturbable input is the tuple $(B_0, \dots, B_k)$, with norm $(\sum_j \|B_j\|_F^2)^{1/2}$, and their random model makes all entries of these coefficient matrices independent complex Gaussians. Thus their main theorem concerns matrix-polynomial eigenvalue conditioning; Remark 1.4 discusses companion algorithms in this setting as well as scalar polynomial rootfinding. The scalar specialization [1, Section 3.1] gives an exact identity for our coefficient ensemble, using the standard Fubini–Study output metric. In our notation,

$$\mu_{\mathrm{FS}}(p, z) = \frac{\|a\|_2 \big(\sum_{k=0}^{n} |z|^{2k}\big)^{1/2}}{(1 + |z|^2)|p'(z)|}, \qquad \mathbb{E}_* \mu_{\mathrm{FS}}^2 = 1.$$

This is relative to the input norm and uses a projective output metric, explained in Section 2.3. It differs from our kernel metric. The companion overlap concerns a further change of input: arbitrary perturbations of C_p. Viewing $zI - C_p$ as a degree-one matrix polynomial does not make the Gaussian coefficient-matrix theorem applicable: its leading matrix is fixed, and C_p has prescribed zeros and ones and ratios of Gaussian coefficients. Our theorem analyzes this induced matrix ensemble. The distinction is therefore between input spaces, metrics, and probability laws, rather than between polynomial solving and eigenvalue problems as such.

Barbarino–Noferini study limiting empirical eigenvalue measures of random matrix polynomials, both with a fixed identity leading coefficient [5] and with random leading coefficients [4]. The latter paper includes our scalar Gaussian ensemble and its limiting uniform root distribution on the unit circle [4, Theorem 1.2]. These results describe root locations; weak convergence of empirical measures does not determine companion overlap moments. Their monic paper also provides bounds for the smallest singular value of a Gaussian matrix with arbitrary deterministic mean [5, Theorem A.5]. These are relevant tools for smoothed analysis, although an application to root conditioning must account for sampling at a root and for the actual perturbation structure. Both papers treat polynomials in one variable; growing matrix size is distinct from a growing number of variables in a polynomial system.

2 The input–output problems

In this section, we specify the coefficient space and local root map, choose the output metric, and define conditioning as a derivative norm. We then identify the matrix perturbation problem and the probability law used to select a root.

2.1 Coefficient space and the local root map

Let $\mathcal{H}$ be an m-dimensional space of complex polynomials on an open set $U \subseteq \mathbb{C}^d$, with a specified Hermitian inner product and orthonormal basis $\phi_1, \dots, \phi_m$. Write $\phi(z) = (\phi_1(z), \dots, \phi_m(z))^T$. The input is a coefficient array $c = (c_{ik}) \in \mathbb{C}^{d\times m}$, representing

$$P_c = (p_1, \dots, p_d)^T, \qquad p_i(z) = \sum_{k=1}^{m} c_{ik}\phi_k(z).$$

We allow perturbations of every coefficient and measure perturbations using the Frobenius norm:

$$\|\delta c\|_F^2 = \sum_{i,k} |\delta c_{ik}|^2.$$

Thus a change of orthonormal coefficient basis changes coordinates, but not the input norm. A nonunitary change of polynomial basis generally changes the problem unless its Gram matrix is included in the norm.

Write $J_c(z) = (\partial p_i/\partial z_j)_{i,j}$ for the Jacobian matrix. A root z of P_c is *simple* if $J_c(z)$ is invertible. We will work with the incidence set

$$\mathcal{V} = \{(c, z) \in \mathbb{C}^{d\times m} \times U : P_c(z) = 0,\ \det J_c(z) \neq 0\}.$$

At each $(c, z) \in \mathcal{V}$, the implicit function theorem gives a unique local root branch

$$\mathcal{S}_{c,z} \colon c' \mapsto z(c'),\ \text{with } z(c) = z. \tag{6}$$

This local map is the input-output map under study. Its derivative at c is

$$D\mathcal{S}_{c,z}[\delta c] = -J_c(z)^{-1} E_z[\delta c], \qquad E_z[\delta c] = \delta c\, \phi(z). \tag{7}$$

Here E_z evaluates the perturbed equations at the original root: $E_z[\delta c] = \delta P(z)$. Derivatives of the root branch and other holomorphic maps are complex-linear.

2.2 Choosing an output metric and defining conditioning

To quantify a change in the output, we must choose how to measure root displacements. Fix a smooth matrix-valued function

$$G : U \longrightarrow \mathbb{C}^{d\times d}, \qquad G(z) = G(z)^*, \qquad v^* G(z) v > 0 \quad (v \neq 0).$$

Thus G assigns a positive definite Hermitian matrix to every possible root location. Identifying the tangent space at $z \in U$ with $\mathbb{C}^d$, define

$$\langle u, v\rangle_{G,z} = u^* G(z) v, \qquad \|v\|_{G,z} = \sqrt{v^* G(z) v}.$$

This field of inner products thus defines a Hermitian metric on U. The constant choice $G(z) = I_d$ gives ordinary Euclidean displacement. Later we will also use a particular nonconstant choice $G = H$, determined by the coefficient space and defined in Section 2.3.

Definition 2.1. For a simple system–root pair (c, z), the *absolute coefficient condition number* with this output metric is defined as

$$\kappa_G(c, z) := \sup_{\delta c \neq 0} \frac{\|D\mathcal{S}_{c,z}[\delta c]\|_{G,z}}{\|\delta c\|_F}. \tag{8}$$

The squared norm of scalar polynomial evaluation at z is

$$S(z) = \|\phi(z)\|_2^2.$$

Thus $S(z) = 0$ if and only if z is a common root of all the basis polynomials $\phi_1, \ldots, \phi_m$, or equivalently of every polynomial in $\mathcal{H}$. For a positive definite Hermitian matrix $G(z)$, the notation $G(z)^{1/2}$ means its unique positive definite Hermitian square root.

The next proposition specializes the standard condition-as-derivative framework [11, Chapter 14] to our coefficient space. We give the short proof to make the evaluation factor and the choice of output metric explicit.

Proposition 2.2. *Fix an output metric G as above and a simple system–root pair (c, z) with $S(z) > 0$. The condition number in (8) is*

$$\kappa_G(c, z) = \sqrt{S(z)}\, \|G(z)^{1/2} J_c(z)^{-1}\|_2. \tag{9}$$

For $G(z) = I_d$, we write $\kappa_{\mathrm{coeff}} := \kappa_{I_d}$, so $\kappa_{\mathrm{coeff}}(c, z) = \sqrt{S(z)}\, \|J_c(z)^{-1}\|_2$. For a fixed holomorphic scalar output $g : U \to \mathbb{C}$, with its usual absolute-value metric, the corresponding condition number is

$$\kappa^g_{\mathrm{coeff}}(c, z) = \sqrt{S}\, \|Dg(z) J^{-1}\|_2. \tag{10}$$

Proof. Suppress c, z temporarily and write $J^{-*} = (J^{-1})^*$. The evaluation map satisfies $E_z E_z^* = S(z) I_d$. Therefore

$$(G^{1/2} J^{-1} E_z)(G^{1/2} J^{-1} E_z)^* = S\, G^{1/2} J^{-1} J^{-*} G^{1/2}.$$

Taking operator norms proves (9) and the chain rule gives (10). □

The assumption $S(z) > 0$ is not needed for Proposition 2.2: if $S(z) = 0$, then $E_z = 0$, so the root derivative and the displayed condition numbers are zero, and the formulas still hold. Positivity will be important later when defining the kernel metric through $\log S$ and normalizing the Jacobian by $S^{-1/2}$.

The distinction between coefficient and residual perturbations is essential. The norm $\|J^{-1}\|_2$ measures sensitivity to perturbing the equation values $P(z)$. Perturbing the coefficient array first evaluates a polynomial at z, which contributes the factor $\sqrt{S}$.

For nonzero input and a nonzero Euclidean output z, the normwise relative condition is

$$\kappa_{\mathrm{coeff,rel}}(c, z) = \kappa_{\mathrm{coeff}}(c, z) \frac{\|c\|_F}{\|z\|_2}. \tag{11}$$

For the scalar output $g(z) \neq 0$, replace the denominator by $|g(z)|$ and use $\kappa^g_{\mathrm{coeff}}$. We do not use the phrase "relative condition" without specifying these input and output scales.

2.3 The kernel metric

We now specify the second output metric used in this paper. The coefficient space determines the reproducing kernel

$$K(z, w) = \sum_{k=1}^{m} \phi_k(z) \overline{\phi_k(w)}, \qquad S(z) = K(z, z).$$

Throughout the general local theory we assume $S > 0$ and

$$H_{jk}(z) = \partial_{\bar{z}_j} \partial_{z_k} \log S(z), \qquad H(z) > 0 \quad (z \in U). \tag{12}$$

The derivatives are taken with z and $\bar{z}$ as independent variables (Wirtinger derivatives). The order of the indices is chosen so that a tangent vector v has squared length $v^* H(z) v$. Taking $G = H$ in the preceding subsection defines the kernel output metric and its condition number κ_H.

There is a geometric interpretation of this formula. The notation $[\phi(z)]$ means the complex line spanned by the nonzero vector $\phi(z)$, a point of complex projective space. The Fubini–Study metric is the standard Hermitian metric on that space of lines. Its *pullback* by $z \mapsto [\phi(z)]$ measures a displacement at z by the displacement it induces in this line; the resulting matrix is $H(z)$. Positivity means that the line changes to first order in every nonzero tangent direction. We use the

normalization whose expression on $\mathbb{P}^1$ is $|\,\mathrm{d}z|^2/(1+|z|^2)^2$. The explicit formula (12) suffices to use the metric without this projective interpretation.

Constructing the root metric from the coefficient inner product is also central to the sparse-system framework of Malajovich–Rojas [21, Sections 1–2]; input normalizations must be matched before comparing condition numbers.

Define the normalized Jacobian

$$A = S^{-1/2} J H^{-1/2}. \tag{13}$$

Proposition 2.2 gives the exact identity

$$\kappa_H(c,z) = \|A^{-1}\|_2. \tag{14}$$

This is a change of output metric, with the coefficient norm held fixed. The Euclidean and kernel-metric condition numbers answer different questions; Section 4 relates them in the box model.

Example 2.3 (Two natural metrics in one variable)**.** For the monomial coefficient norm on polynomials of degree at most n, $S(z) = \sum_{k=0}^{n} |z|^{2k}$. With $r = |z|^2$,

$$H(z) = h_n(r) = \left(r \frac{k_n'(r)}{k_n(r)} \right)', \qquad k_n(r) = \sum_{k=0}^{n} r^k.$$

For the different coefficient norm making $\sqrt{\binom{n}{k}} z^k$ orthonormal, $S(z) = (1+|z|^2)^n$ and $H(z) = n/(1+|z|^2)^2$. The latter is the complex Kostlan model after Gaussian coefficients are introduced. The universal local law below applies to both, but their kernel metrics and their coefficient norms differ.

For comparison with the classical projective theory, consider d homogeneous equations of a common degree n in $d+1$ variables with the Bombieri–Weyl inner product. This makes the monomials $\sqrt{n!/\alpha!}\, X^\alpha$, $|\alpha| = n$, orthonormal, where $\alpha! = \alpha_0! \cdots \alpha_d!$. In the affine chart $X_0 = 1$, their kernel satisfies $S(z) = (1+\|z\|_2^2)^n$, so $H = nG_{\mathrm{FS}}$, with G_{FS} the standard projective metric in this chart. The classical degree-normalized projective condition number is therefore

$$\mu(c,[1:z]) = \|c\|_F\, \kappa_H(c,z). \tag{15}$$

Indeed, at $z = 0$ the normalized projective Jacobian is $J/\sqrt{n}$; unitary invariance gives the identity in every chart [23, 11]. Thus μ includes both an input-norm factor and a degree normalization; the coefficient derivative measured in G_{FS} has norm $\kappa_H/\sqrt{n}$. Neither should be identified with the affine relative condition in (11).

2.4 The matrix eigenvalue problem and its perturbations

For an eigenvalue λ of algebraic multiplicity one of $M \in \mathbb{C}^{N\times N}$, choose nonzero vectors r, w such that $Mr = \lambda r$ and $w^* M = \lambda w^*$. The local eigenvalue map $\mathcal{E}_{M,\lambda}$ sends a nearby matrix to its unique eigenvalue near λ. Its derivative at M is

$$D\mathcal{E}_{M,\lambda}[\delta M] = \frac{w^* \delta M r}{w^* r}.$$

With arbitrary matrix perturbations measured in spectral norm and scalar output measured in absolute value, its condition is

$$\kappa_{\mathrm{eig}}(M,\lambda) = \Lambda(M,\lambda) := \frac{\|w\|_2 \|r\|_2}{|w^* r|} \geq 1. \tag{16}$$

The derivative norm is at most the displayed overlap, and a rank-one perturbation attains it; $\Lambda \geq 1$ follows from Cauchy–Schwarz. The same absolute condition is obtained using Frobenius norm, but its relative version would then use $\|M\|_F$. Here we consistently use

$$\kappa_{\mathrm{eig,rel}}(M,\lambda) = \Lambda(M,\lambda)\frac{\|M\|_2}{|\lambda|} \quad (\lambda \neq 0). \tag{17}$$

Suppose a differentiable algebraic construction $\mathcal{D} : c \mapsto M(c)$ represents $g(z)$ as an eigenvalue. On compatible local branches,

$$\mathcal{E} \circ \mathcal{D} = g \circ \mathcal{S}, \qquad D\mathcal{E}\, D\mathcal{D} = Dg\, D\mathcal{S}. \tag{18}$$

Consequently coefficient-induced matrix perturbations give exactly the original derivative. The comparison in this paper instead allows arbitrary perturbations in the intermediate matrix space. It describes the sensitivity of that larger problem. Errors in constructing M, implementing an eigensolver, and recovering other coordinates require additional analysis.

One important construction uses the quotient algebra of the roots. Suppose the ideal $I = \langle p_1, \ldots, p_d \rangle$ has exactly N distinct complex zeros $z_1, \ldots, z_N$ and consists of all polynomials vanishing at these points (equivalently, I is radical). In $\mathcal{A} = \mathbb{C}[x_1, \ldots, x_d]/I$, two polynomials represent the same element precisely when their values at all roots agree. Evaluation identifies $\mathcal{A}$ with the N-dimensional space of functions on this root set. Choose a basis $\mathcal{B} = (b_1, \ldots, b_N)$ of polynomial representatives. For a polynomial g, multiplication by its class is a linear map on $\mathcal{A}$; denote its matrix, acting on coefficient columns, by M_g. The evaluation matrix $V = (b_k(z_j))_{k,j}$ is invertible, and evaluation gives

$$M_g = V^{-T}\,\mathrm{diag}(g(z_1), \ldots, g(z_N))V^T.$$

If g *separates* the roots, meaning that $g(z_1), \ldots, g(z_N)$ are pairwise distinct, its overlap at z_j is therefore

$$\Lambda_j = \|V^{-T}e_j\|_2\|Ve_j\|_2, \qquad \Lambda_j^2 = \frac{\|Ve_j\|_2^2}{\mathrm{dist}(Ve_j, \mathrm{span}\{Ve_k : k \neq j\})^2}. \tag{19}$$

The last identity follows by orthogonally projecting one evaluation column away from the others. Thus matrix conditioning becomes a question about interpolation in a specified basis norm. This deterministic interpretation also underlies the normal-form analysis in [16].

2.5 Gaussian input and a uniformly selected root

We now make the input random: all dm coefficient coordinates are independent $\mathcal{CN}(0,1)$ variables, with density $\pi^{-1}e^{-|a|^2}$ on $\mathbb{C}$. Equivalently, the real and imaginary parts have variance $1/2$. The components of P are independent and share the kernel K.

Assume that almost surely P has exactly $N \geq 1$ zeros in U, all simple, where N is deterministic. First sample P with this Gaussian law, then select one of its roots uniformly. For a nonnegative measurable function $F(P,z)$, called a *mark*, define

$$\mathbb{E}_* F(P,Z) = \frac{1}{N}\mathbb{E}\sum_{P(z)=0,\ z\in U} F(P,z). \tag{20}$$

This determines a probability law $\mathbb{P}_*$ on system–root pairs. In particular, the marginal law of P is its original Gaussian law. This last fact uses the deterministic root count. No independence between the roots of one system is assumed.

Sampling a root this way is different from conditioning the Gaussian input on $P(z) = 0$ at a prescribed point. The former also weights the conditional law by $|\det J(z)|^2$, as Section 3 proves. We use conditional expectations given $Z = z$ in the regular conditional sense supplied by that formula, rather than treating a zero-probability event as an ordinary event.

For example, for an overlap threshold t,

$$\mathbb{P}_*(\Lambda > t) = \frac{1}{N}\mathbb{E}\#\{j : \Lambda_j > t\}.$$

A tail bound therefore controls the expected fraction of problematic roots. It does not give the same bound for the worst root of a system. Likewise $\mathbb{E}_* \log X$ is a logarithmic average, not a median or a high-probability bound for X.

3 The normalized Jacobian at a selected root

We use the common-kernel Gaussian model and the selected-root law of Section 2.5, with $S > 0$, $H > 0$, and exactly $N \geq 1$ simple roots in U almost surely. The same argument applies to a finite collection of holomorphic functions under these assumptions. Write dV for Lebesgue measure on $\mathbb{C}^d \simeq \mathbb{R}^{2d}$.

3.1 Gaussian regression and the kernel metric

The order of the indices in (12) fixes our complex covariance convention. A derivative $Dp(z)$ is a row, whereas a root displacement is a column. With $D\phi(z)$ the matrix whose kth column is $\partial_{z_k}\phi(z)$, direct differentiation gives

$$H(z) = \frac{1}{S(z)} D\phi(z)^* \left(I - \frac{\phi(z)\phi(z)^*}{S(z)} \right) D\phi(z). \tag{21}$$

Thus $\delta z^* H(z) \delta z$ is the squared norm of the derivative of $z \mapsto [\phi(z)]$ in the Fubini–Study metric, with the normalization specified by (21). Positivity of H says that this projective evaluation map is an immersion.

Lemma 3.1 (Conditional derivative law)**.** *At any fixed $z \in U$,*

$$\mathbb{E}\big[Dp_i(z)^* Dp_i(z) \mid p_i(z) = 0\big] = S(z)H(z).$$

Consequently, conditionally on $P_c(z) = 0$,

$$J_c(z) \overset{\mathrm{d}}{=} \sqrt{S(z)}\, G_0 H(z)^{1/2}, \tag{22}$$

where G_0 has independent $\mathcal{CN}(0,1)$ entries and the square root is the positive Hermitian square root.

Proof. For one component, Gaussian regression applied to (p, Dp) gives

$$\mathbb{E}\big[\overline{\partial_j p}\, \partial_k p \mid p = 0\big] = \partial_{\bar{z}_j}\partial_{z_k} S - \frac{(\partial_{\bar{z}_j} S)(\partial_{z_k} S)}{S} = SH_{jk}.$$

The conditional mean and the conditional pseudo-covariance are zero. The components remain independent after conditioning on their respective values. A standard Gaussian row g satisfies $\mathbb{E}[(gH^{1/2})^*(gH^{1/2})] = H$, which proves (22) with our row convention. □

Recall that $A = S^{-1/2} J H^{-1/2}$ and $\kappa_H = \|A^{-1}\|_2$. Although A is a standard Gaussian matrix under conditioning at a fixed point, its law at a selected root has an additional determinant weight.

3.2 The marked Kac–Rice law

Integration over system–root pairs by coarea is standard in polynomial conditioning; see [7, Sections 5–6] and [2, Remark 5]. The determinant weight below is its complex Kac–Rice form.

Lemma 3.2 (Marked Kac–Rice)**.** *For every nonnegative Borel function F of the full coefficient array and the point,*

$$\mathbb{E} \sum_{z\in U:\, P_c(z)=0} F(c,z) = \int_U \frac{\mathbb{E}\big[F(c,z)|\det J_c(z)|^2 \mid P_c(z)=0\big]}{(\pi S(z))^d}\, dV(z). \tag{23}$$

The equality is valid in $[0,\infty]$. It also holds for signed or complex marks whenever either side with $|F|$ is finite.

Proof. Regard P_c as a smooth Gaussian map between real spaces of dimension $2d$. Its value at z has density $(\pi S(z))^{-d}$ at zero, and holomorphy gives $\det D_{\mathbb{R}} P_c(z) = |\det J_c(z)|^2$. The value distribution is nondegenerate since $S>0$, and the assumed simplicity excludes singular zeros almost surely. The Rice formula therefore applies on compact subsets of U. For a bounded continuous mark, use the full finite coefficient array as the auxiliary Gaussian field in the weighted Rice formula [3, Theorems 6.2 and 6.4]. A monotone-class argument extends the identity to bounded Borel marks. Truncation of the mark and exhaustion of U by compact sets then give (23) by monotone convergence. Finally apply the nonnegative identity to $|F|$ to justify the integrable case. □

Theorem 3.3 (Universal local law)**.** *Under $\mathbb{P}_*$, the root location Z and the normalized Jacobian $A(c,Z)$ are independent. Their densities are respectively*

$$\frac{d!}{N\pi^d}\det H(z), \qquad z\in U, \tag{24}$$

with respect to $dV(z)$, and

$$q_d(A) = \frac{1}{d!\,\pi^{d^2}}|\det A|^2 \exp\!\big(-\|A\|_F^2\big), \qquad A\in\mathbb{C}^{d\times d}, \tag{25}$$

with respect to Lebesgue measure on the matrix entries. Moreover,

$$A^*A \overset{\mathrm{d}}{=} B^*B, \qquad B\in\mathbb{C}^{(d+1)\times d} \tag{26}$$

for B with independent $\mathcal{CN}(0,1)$ entries.

Proof. Apply Lemma 3.2 with $F(c,z)=f(z)g(A(c,z))$, where $f,g\ge 0$. Lemma 3.1 and

$$|\det J_c(z)|^2 = S(z)^d \det H(z)\,|\det A(c,z)|^2$$

yield

$$\mathbb{E}_*[f(Z)g(A)] = \frac{1}{N\pi^d}\int_U f(z)\det H(z)\, dV(z)\mathbb{E}\big[g(G_0)|\det G_0|^2\big]. \tag{27}$$

Expanding the determinant over permutations gives $\mathbb{E}|\det G_0|^2 = d!$: only equal pairs of permutations have nonzero expectation. With $f=g=1$, (27) shows that (24) integrates to one. The factorization then proves independence and (25).

A complex Wishart matrix B^*B, with B of size $m\times d$, has density proportional to $(\det W)^{m-d}e^{-\operatorname{tr} W}$ on positive Hermitian matrices. The square Gaussian Gram matrix $G_0^*G_0$ has exponent zero. Multiplication of its density by $|\det G_0|^2 = \det(G_0^*G_0)$ therefore gives precisely the case $m=d+1$, proving (26). □

The matrix law (25) is the complex induced Ginibre ensemble with parameter one [15]. The zero-density formula belongs to the established covariance-kernel approach to Gaussian zeros [9, 21]. These are standard ingredients. Their role here is to identify exactly the distribution of the coefficient-to-root derivative in the kernel metric, separating it from the distribution of root locations and from any algebraic representation.

3.3 A sharp tail for local conditioning

Theorem 3.4 (Tail and moments)**.** *Put $C_d = d(d+1)/4$. For every $t > 0$,*

$$\mathbb{P}_*\{\kappa_H > t\} \le \min\{1, C_d t^{-4}\}, \qquad \lim_{t\to\infty} t^4 \mathbb{P}_*\{\kappa_H > t\} = C_d. \tag{28}$$

In particular, for $q > 0$,

$$\mathbb{E}_*\kappa_H^q < \infty \quad \Longleftrightarrow \quad q < 4, \qquad \mathbb{E}_*\kappa_H^q \le \frac{4}{4-q} C_d^{q/4} \quad (0 < q < 4). \tag{29}$$

The inverse Frobenius moment is exactly

$$\mathbb{E}_*\|A^{-1}\|_F^2 = d. \tag{30}$$

Also $\mathbb{E}_*|\log \kappa_H| < \infty$.

Proof. Let $X = \sigma_{\min}(A)^2$. By (26), this is the smallest eigenvalue of a complex Wishart matrix with rectangularity one. Its density is [17, Section 2.2, equation (15)]

$$f_d(x) = xe^{-dx} L_{d-1}^{(2)}(-x) = xe^{-dx} \sum_{k=0}^{d-1} \binom{d+1}{k+2} \frac{x^k}{k!}, \qquad x > 0. \tag{31}$$

For $0 \le k \le d-1$,

$$\frac{\binom{d+1}{k+2}}{\binom{d+1}{2}} = \frac{2(d-1)(d-2)\cdots(d-k)}{(k+2)!} \le d^k,$$

where the empty product is one. Hence $f_d(x) \le \binom{d+1}{2} x$. Since $\kappa_H = X^{-1/2}$, integration over $0 < x < t^{-2}$ proves the bound in (28). The expansion $f_d(x) = \binom{d+1}{2} x + O_d(x^2)$ at zero proves its sharp asymptotic. Integrating the tail gives both the moment threshold and, on splitting at $t = C_d^{1/4}$, the bound in (29).

There is an elementary cofactor proof of (30):

$$\mathbb{E}_*\|A^{-1}\|_F^2 = \frac{\mathbb{E}[|\det G_0|^2 \|G_0^{-1}\|_F^2]}{d!} = \frac{\mathbb{E}\|\operatorname{adj} G_0\|_F^2}{d!} = \frac{d^2(d-1)!}{d!} = d.$$

Each cofactor is the determinant of a $(d-1)\times(d-1)$ standard complex Gaussian matrix and has second moment $(d-1)!$; at $d = 1$ the empty determinant is one. Finally, the positive logarithmic part follows from any positive moment below four. For the negative part, use $\log^+(1/\kappa_H) = \log^+ \sigma_{\min}(A) \le \log^+ \|A\|_F$ and the Gaussian decay in (25). □

Two consequences clarify the scope of the theorem. First,

$$\mathbb{P}_*\{\kappa_H > \sqrt{d}\, t\} \le \min\{1, (2t^4)^{-1}\}. \tag{32}$$

Thus the geometric normalization gives a tail uniform in the coefficient space, polynomial degrees, and, after the scale $\sqrt{d}$, the dimension. Second, (20) makes the same probability an expected fraction of roots:

$$\frac{1}{N}\mathbb{E}\#\{z \in U : P_c(z) = 0,\ \kappa_H(c,z) > t\} = \mathbb{P}_*\{\kappa_H > t\}.$$

This statement does not assume independence between different roots.

3.4 Comparison with classical projective moments

In the common-degree Bombieri–Weyl model of (15), the classical Frobenius variant is $\mu_F(c,[1:z]) = \|c\|_F\|A^{-1}\|_F$. Consequently (30) recovers the centered identity

$$\mathbb{E}_* \frac{\mu_F(c,[1:Z])^2}{\|c\|_F^2} = d \tag{33}$$

of Armentano–Beltrán–Bürgisser–Cucker–Shub [2, Theorem 2]. Their theorem also allows unequal degrees and gives the bound d/σ^2 for Gaussian perturbations of an arbitrary deterministic homogeneous system, with equality at zero center. Thus this homogeneous second-moment identity is already known.

Beltrán–Pardo [7, Sections 7–8] give exact operator moments of μ for orders $0 < q < 4$ under spherical Bombieri–Weyl input. Those results use the same rectangular Gaussian singular-value mechanism. Malajovich–Rojas [21, Theorem 1] already obtain a fourth-power tail for the event that some sparse-system root has large condition number, with their projective input normalization. Theorem 3.4 gives the explicit distribution and sharp constant for the absolute kernel-metric condition at a selected root under our common-kernel assumptions. Its moment threshold is consistent with these earlier results; it does not establish a new exponent for polynomial systems.

Example 3.5 (One complex equation)**.** When $d = 1$, the variable $|A|^2$ has density xe^{-x} on $x > 0$. For every nondegenerate finite-dimensional Gaussian polynomial ensemble covered by the hypotheses,

$$\mathbb{P}_*\{\kappa_H > t\} = 1 - e^{-t^{-2}}(1 + t^{-2}), \qquad \mathbb{E}_*\kappa_H^2 = 1.$$

The tail is asymptotic to $1/(2t^4)$, independently of the degree.

There is also a short consistency check for the known projective identity $\mathbb{E}_*\mu_{\mathrm{FS}}^2 = 1$ in the scalar monomial model [1]. Kac–Rice cancels the derivative denominator, while the conditional coefficient vector given $p(z) = 0$ is a standard Gaussian on an n-dimensional hyperplane. Hence

$$\mathbb{E}_*\mu_{\mathrm{FS}}^2 = \frac{1}{n\pi}\int_{\mathbb{C}} \frac{\mathbb{E}[\|a\|_2^2 \mid p(z) = 0]}{(1+|z|^2)^2}\, dV(z) = \frac{1}{\pi}\int_{\mathbb{C}} \frac{dV(z)}{(1+|z|^2)^2} = 1.$$

This identity and Example 3.5 concern different condition numbers, despite their equal second moments.

4 The box ensemble: root location and coefficient scale

Fix $d, n \geq 1$ and let

$$p_i(z) = \sum_{\alpha \in \{0,\dots,n\}^d} c_{i,\alpha} z^\alpha, \qquad c_{i,\alpha} \sim \mathcal{CN}(0,1) \quad \text{independently.} \tag{34}$$

The monomials are orthonormal in the coefficient norm. For generic coefficients, the system has

$$N = d!n^d \tag{35}$$

simple affine roots, all in $(\mathbb{C}^*)^d$. This is the mixed-volume count for d identical boxes $[0,n]^d$ [8, 13]. The exceptional coefficient sets are proper algebraic sets and hence have Gaussian measure zero. Equivalently, the multihomogeneous compactification on $(\mathbb{P}^1)^d$ has intersection number $d!n^d$, with no

boundary intersection for generic sections. This statement concerns that compactification; ordinary homogenization to total degree dn can introduce common zeros at projective infinity.

Put

$$k_n(r) = \sum_{j=0}^{n} r^j, \qquad h_n(r) = \left(r \frac{k_n'(r)}{k_n(r)} \right)'.$$

The covariance kernel and its metric are

$$K(z,w) = \prod_{\ell=1}^{d} k_n(z_\ell \bar{w}_\ell), \qquad H(z) = \operatorname{diag}(h_n(|z_1|^2), \dots, h_n(|z_d|^2)). \tag{36}$$

In particular $H > 0$. A convenient interpretation is obtained by giving $J_r \in \{0, \dots, n\}$ probabilities $\mathbb{P}(J_r = j) = r^j / k_n(r)$. Differentiation gives

$$r h_n(r) = \operatorname{Var}(J_r), \qquad r > 0. \tag{37}$$

The continuous endpoint values are $h_n(0) = 1$ and $h_n(1) = n(n+2)/12$.

4.1 An exact selected-root law

Proposition 4.1 (Independent radial coordinates)**.** *Under $\mathbb{P}_*$, the coordinates of Z are independent, their angles are independent and uniform, and $R_\ell = |Z_\ell|^2$ has distribution*

$$F_n(r) = \mathbb{P}_*(R_\ell \le r) = \frac{r k_n'(r)}{n k_n(r)}, \qquad F_n'(r) = \frac{h_n(r)}{n} \quad (r \ge 0). \tag{38}$$

Moreover, R_ℓ and R_ℓ^{-1} have the same law.

Proof. Theorem 3.3 and (35) give joint density $\prod_\ell h_n(|z_\ell|^2)/(n\pi)$ with respect to complex Lebesgue measure. In polar coordinates $\mathrm{d}V(z_\ell) = \frac{1}{2}\,\mathrm{d}r_\ell\,\mathrm{d}\theta_\ell$, which proves the density statement. Integration gives (38), with limits zero and one since $r k_n'/k_n$ tends to zero and n at the two endpoints. The identity $k_n(r) = r^n k_n(r^{-1})$ implies $F_n(r) = 1 - F_n(r^{-1})$ for $r > 0$, proving inversion symmetry. □

The independence in this proposition is between coordinates of one selected root. Different roots of the same polynomial system remain dependent.

Lemma 4.2 (Depth and the outer-root moment)**.** *For any one coordinate, put $s = n \log R$. Then*

$$\mathbb{P}_*(|s| > t) \le \min\{1, 2/t\} \quad (t > 0), \qquad \mathbb{E}_* \log^+ R = \frac{\log(n+1)}{n}. \tag{39}$$

Consequently $\sup_n \mathbb{E}_* \log(1 + |s|) < \infty$.

Proof. For $0 < r < 1$, the mean of the truncated geometric law in (37) is at most the untruncated mean $r/(1-r)$. Thus

$$F_n(e^{-t/n}) \le \frac{1}{n(e^{t/n} - 1)} \le \frac{1}{t}.$$

Inversion symmetry gives the two-sided tail. Also

$$\begin{aligned} \mathbb{E}_* \log^+ R &= \int_1^\infty \frac{1 - F_n(r)}{r}\,\mathrm{d}r \\ &= \frac{1}{n}\left[\log \frac{r^n}{k_n(r)} \right]_{r=1}^{r=\infty} = \frac{\log(n+1)}{n}. \end{aligned}$$

Finally integrate the depth tail against $(1+t)^{-1}\,\mathrm{d}t$. □

4.2 Uniform metric estimates

The next elementary estimate is used both for coefficient conditioning and for the companion overlap. It replaces a boundary-layer asymptotic by a bound valid at every radius and degree.

Lemma 4.3 (The one-variable kernel metric)**.** *There are absolute constants $0 < c < C < \infty$ such that, for $n \geq 1$ and $0 \leq r \leq 1$,*

$$\frac{c}{(1 - r + 1/n)^2} \leq h_n(r) \leq \frac{C}{(1 - r + 1/n)^2}. \tag{40}$$

For $r > 0$,

$$h_n(r) = r^{-2} h_n(r^{-1}). \tag{41}$$

Proof. For $0 \leq r \leq 1/4$, the pair of indices $0, 1$ in the variance formula gives $h_n(r) \geq k_n(r)^{-2} \geq (1 - r)^2 \geq 9/16$. Direct differentiation of $k_n(r) = (1 - r^{n+1})/(1 - r)$ gives, for $0 < r < 1$,

$$h_n(r) = \frac{1}{(1 - r)^2} - \frac{(n + 1)^2 r^n}{(1 - r^{n+1})^2} \leq \frac{1}{(1 - r)^2}. \tag{42}$$

Thus h_n is bounded above and below by absolute positive constants on $[0, 1/4]$, which proves the assertion there.

For $1/4 \leq r < 1$, set

$$m = \min\{n, \lfloor (1 - r)^{-1} \rfloor\}.$$

Then $k_n(r) \leq m + 1$. Also $\log r \geq -(1 - r)/r \geq -4(1 - r)$, so $r^j \geq e^{-4}$ for $0 \leq j \leq m$. Each of these indices has probability at least $e^{-4}/(m + 1)$. Using an independent copy J_r',

$$\operatorname{Var}(J_r) = \frac{1}{2}\mathbb{E}(J_r - J_r')^2 \geq e^{-8} \operatorname{Var}(\operatorname{Unif}\{0, \ldots, m\}) = e^{-8} \frac{m(m + 2)}{12}.$$

Since $m \geq \frac{1}{2} \min\{n, (1 - r)^{-1}\}$, division by $r \leq 1$ proves the lower bound in (40). For the upper bound, $\operatorname{Var}(J_r) \leq n^2/4$ and $r \geq 1/4$ imply $h_n(r) \leq n^2$. Combining this with (42) gives

$$h_n(r) \leq \min\{n^2, (1 - r)^{-2}\} \leq \frac{4}{(1 - r + 1/n)^2}.$$

At $r = 1$, use $h_n(1) = n(n + 2)/12$. Finally, differentiating $F_n(r) = 1 - F_n(r^{-1})$ proves (41). □

Corollary 4.4 (Logarithmic metric scale)**.** *For R with distribution* (38)*,*

$$\mathbb{E}_* \left| \log \frac{h_n(R)}{n^2} \right| \leq C \tag{43}$$

with an absolute constant. In particular $\mathbb{E}_* \log h_n(R) = 2 \log n + O(1)$.

Proof. For $0 < r \leq 1$, write $s = n \log r \leq 0$. Since $n(1 - r) \leq |s|$, Lemma 4.3 gives

$$\frac{c}{(1 + |s|)^2} \leq \frac{h_n(r)}{n^2} \leq C.$$

For $r > 1$, apply reversal. At every $r > 0$ this yields

$$\left| \log \frac{h_n(r)}{n^2} \right| \leq C + 2 \log(1 + |n \log r|) + 2 \log^+ r.$$

Take expectations and apply Lemma 4.2, noting that $\log(n + 1)/n \leq \log 2$. □

4.3 Coefficient conditioning and representation overlap

Proposition 4.5 (The coefficient scale)**.** *For fixed d in the box model,*

$$\mathbb{E}_* \log \kappa_{\mathrm{coeff}}(P, Z) = -\log n + O_d(1). \tag{44}$$

The same formula holds for the matched scalar coordinate condition $\kappa^{x_1}_{\mathrm{coeff}}$ in (10)*. All logarithms here are absolutely integrable for each n, d.*

Proof. Put $H = n^2 D$, with diagonal entries $h_n(R_\ell)/n^2$. From $J = \sqrt{S} A H^{1/2}$ we obtain

$$\kappa_{\mathrm{coeff}} = n^{-1} \|D^{-1/2} A^{-1}\|_2.$$

The logarithm of $\|D^{-1/2}A^{-1}\|_2$ differs from that of $\|A^{-1}\|_2$ by at most $\frac{1}{2}\max_\ell |\log D_{\ell\ell}|$. Theorem 3.4 and Corollary 4.4 give (44) and absolute integrability. For the scalar output there is the more precise factorization

$$\kappa^{x_1}_{\mathrm{coeff}} = h_n(R_1)^{-1/2} \|e_1^T A^{-1}\|_2. \tag{45}$$

The root location and A are independent. The row norm has an integrable logarithm because it is at most $\|A^{-1}\|_2$ and at least $\|A\|_2^{-1}$. Apply Corollary 4.4 again. □

Suppose now that a fixed monomial set $\mathcal{B}$ is a quotient basis on a generic coefficient chart. Use multiplication by x_1, whose root values are distinct almost surely in the box model. One way to check the last assertion is to consider pairs $z \neq w$ with $z_1 = w_1$: evaluation at these points gives two independent linear conditions on each polynomial, since constants and all coordinates belong to the coefficient space. The pair incidence set has dimension at most one less than coefficient space. Its projection therefore has Gaussian measure zero.

Define the extended nonnegative expectation $\Theta^{\mathcal{B}}_{n,d} = \mathbb{E}_* \log \Lambda_{\mathcal{B}}$, where the overlap uses the Euclidean norm on the chosen quotient coordinates. Equations (16) and (44) imply

$$\mathbb{E}_* \log \frac{\kappa_{\mathrm{eig}}(M_{x_1}, Z_1)}{\kappa^{x_1}_{\mathrm{coeff}}(P, Z)} = \log n + \Theta^{\mathcal{B}}_{n,d} + O_d(1). \tag{46}$$

This identity also holds with the full-root condition in the denominator, but (46) compares the same scalar output. If $\Theta^{\mathcal{B}}_{n,d} = +\infty$, interpret the equality in the extended sense; the other logarithmic terms are integrable. No bound on multivariate overlap is asserted here. Section 5 controls it for the univariate companion basis.

For completeness, the evaluation factor itself satisfies

$$\mathbb{E}_* \log k_n(R) = 2\log n + O(1), \qquad \mathbb{E}_* \log \sqrt{S(Z)} = d \log n + O_d(1). \tag{47}$$

Indeed reversal gives the exact identity

$$\log k_n(R) = \log k_n(e^{-|s|/n}) + s_+.$$

Here $\mathbb{E}_* s_+ = \log(n+1)$, while

$$\frac{n}{1+|s|} \leq k_n(e^{-|s|/n}) \leq n+1.$$

The lower bound follows by comparing the sum with $\int_0^n e^{-|s|x/n}\,\mathrm{d}x$ and using $1 - e^{-u} \geq u/(1+u)$. Lemma 4.2 proves (47). Replacing coefficient conditioning by residual conditioning would therefore change the leading term in (46) from $\log n$ to $(d+1)\log n$. The outer roots contribute to this difference, even though most roots lie close to the unit polytorus.

5 Conditioning of companion eigenvalues

We now keep the scalar output fixed and change its representation. Let

$$p(x) = \sum_{j=0}^{n} a_j x^j, \qquad a_0, \dots, a_n \text{ independent with law } \mathcal{CN}(0,1),$$

and let C_p be multiplication by x on $\mathbb{C}[x]/(p)$ in the ordered basis $1, x, \dots, x^{n-1}$. Thus C_p has ones on its subdiagonal and last column $-(a_0, \dots, a_{n-1})^T/a_n$. Almost surely $a_n \neq 0$ and all roots are simple. For a root z, put

$$v(z) = (1, z, \dots, z^{n-1})^T, \qquad \ell_z(x) = \frac{p(x)}{(x-z)p'(z)}.$$

The coefficient vector of ℓ_z is a right eigenvector of C_p, whereas $v(z)^T$ is a left eigenvector, and $\ell_z(z) = 1$. Consequently the absolute eigenvalue condition number for spectral-norm matrix perturbations is

$$\Lambda(p, z) = \|v(z)\|_2 \, \|\ell_z\|_2. \tag{48}$$

Here the second norm is the Euclidean norm of the coefficient vector in the displayed monomial basis. In particular $\Lambda \geq 1$. This basis choice is part of the theorem: the overlap is not invariant under a general change of polynomial basis.

The next theorem proves part (i) of Theorem 1.1. The comparison in part (ii) follows in Corollary 5.7.

Theorem 5.1 (Uniform companion-overlap tails)**.** *There is an absolute constant C such that, for every $n \geq 1$ and $t > 0$,*

$$\mathbb{P}_*\{\Lambda(p, Z) > t\} = \frac{1}{n}\mathbb{E}\#\{z : p(z) = 0,\ \Lambda(p, z) > t\} \leq \min\{1, Ct^{-2}\}. \tag{49}$$

For every $0 < q < 2$ there is a constant C_q, independent of n, with

$$\mathbb{E}_*\Lambda^q \leq C_q, \qquad 0 \leq \mathbb{E}_* \log \Lambda \leq C.$$

The moment range is sharp for bounds uniform in the degree:

$$\mathbb{E}_*\Lambda^2 = 2\log n + O(1) \qquad (n \to \infty). \tag{50}$$

For $n = 1$, the overlap is identically one.

The tail controls the expected fraction of poorly conditioned eigenvalues. It does not require independence of the roots and does not assert a degree-independent bound for the worst root of a polynomial. The second-moment assertion also shows that the exponent 2 in (49) cannot be replaced by any larger exponent with a degree-independent constant.

5.1 An exact conditional moment

Write $k_j(r) = \sum_{i=0}^{j} r^i$ and $h_n(r) = (rk_n'(r)/k_n(r))'$, as in the box model. At a fixed location z, coefficient multiplication by $x - z$ is the injective map

$$D_z : \mathbb{C}^n \longrightarrow \mathbb{C}^{n+1}, \qquad \mathbf{q} \longmapsto \operatorname{coeff}((x-z)q(x)).$$

Its Gram matrix $T_z = D_z^* D_z$ is tridiagonal, with diagonal $1 + r$, superdiagonal $-z$, and subdiagonal $-\overline{z}$, where $r = |z|^2$. In particular it is positive definite, including at $z = 0$.

Proposition 5.2 (Conditional overlap identity)**.** *Under the selected-root law, the conditional second moment is*

$$\Phi_n(r) := \mathbb{E}_*[\Lambda^2 \mid Z = z] = \frac{k_{n-1}(r)}{k_n(r)h_n(r)} \operatorname{tr}(T_z^{-1}) = \frac{k_{n-1}(r)A_n(r)}{k_n(r)^2 h_n(r)}, \tag{51}$$

where

$$A_n(r) = \sum_{j=0}^{n-1} (j+1)(n-j)r^j. \tag{52}$$

The identity defines a continuous version of the conditional moment. It satisfies

$$\Phi_n(r) = \Phi_n(r^{-1}) \quad (r > 0), \qquad \Phi_n(0) = n, \qquad \Phi_n(1) = \frac{2n}{n+1}.$$

For $n = 1$, $\Phi_1(r) = 1$ for all $r \geq 0$.

Proof. Ordinary Gaussian conditioning on $p(z) = 0$ restricts the standard Gaussian coefficient vector to the hyperplane $\operatorname{im} D_z$. Writing $p(x) = (x - z)q(x)$ therefore gives

$$\mathbf{q} \mid \{p(z) = 0\} \sim \mathcal{CN}(0, T_z^{-1}).$$

Indeed its density in these coordinates is proportional to $\exp(-\mathbf{q}^* T_z \mathbf{q})$; the Jacobian of the linear parametrization is constant. Gaussian regression also gives

$$\mathbb{E}[|p'(z)|^2 \mid p(z) = 0] = k_n(r)h_n(r).$$

At a selected root, the conditional law is tilted by $|p'(z)|^2$. Since $p'(z) = q(z)$ and $\Lambda^2 = k_{n-1}(r)\|\mathbf{q}\|_2^2/|p'(z)|^2$, the tilt cancels the denominator. Thus

$$\mathbb{E}_*[\Lambda^2 \mid Z = z] = \frac{k_{n-1}(r)\mathbb{E}[\|\mathbf{q}\|_2^2 \mid p(z) = 0]}{\mathbb{E}[|p'(z)|^2 \mid p(z) = 0]},$$

which proves the first equality in (51).

The determinant of the $j \times j$ leading block of T_z is $k_j(r)$: both satisfy the recurrence $d_j = (1 + r)d_{j-1} - rd_{j-2}$, with $d_0 = 1$ and $d_1 = 1 + r$. Deleting its ith row and column leaves two such blocks, so

$$\operatorname{tr}(T_z^{-1}) = \frac{1}{k_n(r)} \sum_{i=1}^{n} k_{i-1}(r)k_{n-i}(r) = \frac{A_n(r)}{k_n(r)}. \tag{53}$$

The coefficient of r^j in the numerator is $(j + 1)(n - j)$, which proves the second equality.

Reversal provides a useful check and removes the distinction between inner and outer roots. For $z \neq 0$, let $p^{\mathrm{rev}}(x) = x^n p(1/x)$. Its coefficients have the same law as those of p, and direct reversal of ℓ_z gives

$$\|\ell_{1/z}^{\mathrm{rev}}\|_2 = |z|^{n-1}\|\ell_z\|_2, \qquad \|v(1/z)\|_2 = |z|^{-(n-1)}\|v(z)\|_2.$$

Hence $\Lambda(p^{\mathrm{rev}}, 1/z) = \Lambda(p, z)$. Reversal carries a uniformly selected root to a uniformly selected root, proving the symmetry of Φ_n. Finally,

$$h_n(1) = \frac{n(n+2)}{12}, \qquad \operatorname{tr}(T_1^{-1}) = \frac{n(n+2)}{6}.$$

Substitution proves the values at $r = 1$; the assertions at $r = 0$ and $n = 1$ follow directly from (51). □

5.2 Uniform estimates and a tail bound

For $0 \le r \le 1$, the cofactor formula (53) immediately gives

$$\operatorname{tr}(T_z^{-1}) \le n \min\{n, (1-r)^{-1}\} \le \frac{2n}{1-r+1/n}, \tag{54}$$

with the usual interpretation at $r = 1$. Combining this with the metric estimate (40), $k_{n-1} \le k_n$, and reversal proves

$$\Phi_n(r) \le C(1 + n(1 - \min\{r, r^{-1}\})) \le C(1 + |n \log r|), \qquad r > 0. \tag{55}$$

The conditional second moment alone gives all the uniform moments of order below two. A little more of the same Gaussian regression gives the sharper tail in Theorem 5.1.

Lemma 5.3 (Conditional moments)**.** *For each $0 < q < 4$ there is a constant C_q, independent of n and z, such that*

$$\mathbb{E}_*[\Lambda^q \mid Z = z] \le C_q \Phi_n(|z|^2)^{q/2}.$$

Proof. Under ordinary conditioning on $p(z) = 0$, put $\Sigma = T_z^{-1}$ and $\sigma^2 = v(z)^T \Sigma \overline{v(z)} = k_n(r) h_n(r)$. Gaussian regression decomposes the coefficient vector as

$$\mathbf{q} = b\xi + \eta, \qquad b = \Sigma\overline{v(z)}/\sigma, \qquad \xi = q(z)/\sigma \sim \mathcal{CN}(0,1),$$

where η is independent of ξ and has covariance $R = \Sigma - bb^*$. At a selected root, the tilt is $|\xi|^2$. It preserves the law of η and its independence from ξ, while $|\xi|^2$ now has density ue^{-u} on $(0,\infty)$. In particular

$$\mathbb{E}_*|\xi|^{-q} = \int_0^\infty u^{1-q/2} e^{-u}\, du = \Gamma(2 - q/2) < \infty, \qquad 0 < q < 4.$$

Writing $\beta = \|v(z)\|_2/\sigma$, we have

$$\Lambda = \beta \frac{\|b\xi + \eta\|_2}{|\xi|} \le \beta\|b\|_2 + \beta \frac{\|\eta\|_2}{|\xi|}, \qquad \Phi_n(r) = \beta^2 \operatorname{tr}\Sigma.$$

The standard Gaussian fourth-moment identity, valid also for a degenerate covariance, is

$$\mathbb{E}\|\eta\|_2^4 = (\operatorname{tr} R)^2 + \operatorname{tr}(R^2) \le 2(\operatorname{tr} R)^2.$$

It follows by diagonalizing R and using $\mathbb{E}|g|^2 = 1$ and $\mathbb{E}|g|^4 = 2$ for a standard complex Gaussian g. Hence $\mathbb{E}\|\eta\|_2^q \le 2^{q/4}(\operatorname{tr} R)^{q/2}$ for $0 < q < 4$. Apply $(a+b)^q \le 2^{\max\{q-1,0\}}(a^q + b^q)$, independence, and $\|b\|_2^2 \le \operatorname{tr}\Sigma$, $\operatorname{tr} R \le \operatorname{tr}\Sigma$ to obtain the claim. □

Proposition 5.4 (Fixed-degree integrability)**.** *For each fixed $n \ge 2$ and $q > 0$,*

$$\mathbb{E}_*\Lambda^q < \infty \quad \Longleftrightarrow \quad q < 4.$$

For $n = 1$ every moment is finite, since $\Lambda = 1$.

Proof. The stronger first bound in (55) gives $\Phi_n(r) \le Cn$ at every radius. Lemma 5.3 therefore proves finiteness for $q < 4$. For the converse, the residual covariance R in its proof has rank $n-1$: regression removes one complex direction from the positive definite covariance Σ. If $n \ge 2$, there is thus an $a > 0$ with $\mathbb{P}\{\|\eta\|_2 > a\} > 0$. On this event, for sufficiently small $|\xi|$, one has $\|b\xi + \eta\|_2 > a/2$. Independence and the density ue^{-u} of $|\xi|^2$ then show that $\mathbb{E}_*[\Lambda^q \mid Z = z] = \infty$ for $q \ge 4$, because $\int_0^\varepsilon u^{1-q/2} e^{-u}\, du = \infty$. Averaging over Z proves the assertion. □

Thus the threshold four for each fixed degree comes from the small derivative direction in the root-conditioned Gaussian. The lower threshold two for bounds uniform in degree also reflects the distribution of root locations.

Proof of the tail and moment assertions in Theorem 5.1. Let $D = |n \log |Z|^2|$ and $W = 1 + D$. The box root law (39) gives $\mathbb{P}_*\{D > u\} \le \min\{1, 2/u\}$, so $\mathbb{P}_*\{W > u\} \le \min\{1, 4/u\}$. By conditional Markov, Lemma 5.3 with $q = 3$, and (55),

$$\mathbb{P}_*\{\Lambda > t\} \le \mathbb{E}_* \min\{1, CW^{3/2}/t^3\}.$$

For every $A > 0$, integration of the tail gives

$$\mathbb{E}_* \min\{1, (W/A)^{3/2}\} = \frac{3}{2A^{3/2}} \int_0^A u^{1/2} \mathbb{P}_*\{W > u\}\, du \le \min\{1, C/A\}.$$

Taking A to be a constant multiple of t^2 proves (49). Tail integration then proves $\sup_n \mathbb{E}_* \Lambda^q < \infty$ for $0 < q < 2$ and $\sup_n \mathbb{E}_* \log \Lambda < \infty$. Nonnegativity of the logarithm follows from $\Lambda \ge 1$. □

5.3 The critical second moment

Proposition 5.5. *For every $n \ge 1$,*

$$\mathbb{E}_* \Lambda^2 = \frac{2}{n} \int_0^1 \frac{k_{n-1}(r) A_n(r)}{k_n(r)^2}\, dr. \tag{56}$$

In particular $\mathbb{E}_* \Lambda^2 = 2 \log n + O(1)$ as $n \to \infty$.

Proof. The density of $|Z|^2$ is $h_n(r)/n$. It cancels the h_n in (51), and reversal makes the contributions of $(0, 1)$ and $(1, \infty)$ equal. This proves (56), including finiteness for each n.

For $0 \le r < 1$, summing (52) gives

$$A_n(r) = \frac{n - (n+2)r + (n+2)r^{n+1} - nr^{n+2}}{(1-r)^3}.$$

If $G_n(r) = k_{n-1}(r) A_n(r)/(n k_n(r)^2)$, rearrangement yields

$$G_n(r) = \frac{1}{1-r} \frac{(1-r^n)(1+r^{n+1})}{(1-r^{n+1})^2} - \frac{2r}{n(1-r)^2} \left(\frac{1-r^n}{1-r^{n+1}} \right)^2.$$

Suppose $n \ge 2$. On $0 \le r \le 1 - 1/n$ we have $r^{n+1} \le e^{-1}$, and therefore

$$\left| G_n(r) - \frac{1}{1-r} \right| \le \frac{C r^n}{1-r} + \frac{2}{n(1-r)^2}.$$

The right side has a uniformly bounded integral on this interval: for its first term use $r^n \le e^{-n(1-r)}$ and $u = n(1-r) \ge 1$; the second term integrates directly. Hence

$$\int_0^{1-1/n} G_n(r)\, dr = \log n + O(1).$$

On the remaining interval, (54) and $k_{n-1} \le k_n$ give $0 \le G_n(r) \le n$, so its integral is at most one. Multiplication by two proves the asserted asymptotic. □

This completes Theorem 5.1. By monotonicity of moments on a probability space, $\sup_n \mathbb{E}_* \Lambda^q = \infty$ for every $q \geq 2$, allowing the value $+\infty$ for a fixed degree. Thus the sharpness assertion concerns uniformity in n; it does not say that all moments of order at least two diverge at each fixed degree.

Example 5.6 (Degrees one and two)**.** When $n = 1$, C_p is a scalar and $\Lambda = 1$. When $n = 2$,

$$T_z = \begin{pmatrix} 1+r & -z \\ -\overline{z} & 1+r \end{pmatrix}, \qquad \Phi_2(r) = \frac{2(1+r)^2}{1+4r+r^2}.$$

In particular the conditional second moment is $4/3$ on the unit circle, and direct integration of (56) gives

$$\mathbb{E}_* \Lambda^2 = \frac{2}{3} + \frac{4\pi}{9\sqrt{3}}.$$

The deterministic overlap can nevertheless become arbitrarily large: if $p(x) = (x-z)(x-w)$ with $z \neq w$, then $\Lambda(p, z) = \sqrt{(1+|z|^2)(1+|w|^2)}/|z-w|$. The theorem describes how averaging over the polynomial and a selected root controls these near-collision configurations.

5.4 Comparison with coefficient conditioning

For the same scalar output z, coefficient and matrix perturbations give, respectively,

$$\kappa_{\mathrm{coeff}}(p, z) = \frac{\sqrt{k_n(|z|^2)}}{|p'(z)|}, \qquad \kappa_{\mathrm{eig}}(C_p, z) = \Lambda(p, z).$$

All $n+1$ polynomial coefficients are perturbable in the first condition number, including the leading coefficient. Multiplication by the corresponding data norm and division by the common output norm $|z|$ define their relative versions; $z \neq 0$ almost surely.

Corollary 5.7 (A relative conditioning comparison)**.** *With Euclidean coefficient norm and spectral matrix norm, define*

$$\kappa_{\mathrm{coeff,rel}}(p, z) = \frac{\sqrt{k_n(|z|^2)}\|\mathbf{a}\|_2}{|p'(z)|\,|z|}, \qquad \kappa_{\mathrm{eig,rel}}(C_p, z) = \frac{\Lambda(p, z)\|C_p\|_2}{|z|}.$$

Then

$$\mathbb{E}_* \log \frac{\kappa_{\mathrm{eig}}}{\kappa_{\mathrm{coeff}}} = \log n + O(1),$$
$$\mathbb{E}_* \log \frac{\kappa_{\mathrm{eig,rel}}}{\kappa_{\mathrm{coeff,rel}}} = \log n + O(1).$$

The error terms are bounded uniformly in the degree.

Proof. The normalized derivative $p'(Z)/\sqrt{k_n(|Z|^2)h_n(|Z|^2)}$ has the determinant-weighted complex Gaussian law from the local theorem: its squared modulus has density ue^{-u}. Consequently the following finite-degree identity holds, where γ is Euler's constant:

$$\mathbb{E}_* \log \frac{\kappa_{\mathrm{eig}}}{\kappa_{\mathrm{coeff}}} = \mathbb{E}_* \log \Lambda + \frac{1}{2}\mathbb{E}_* \log h_n(|Z|^2) + \frac{1-\gamma}{2}. \tag{57}$$

The bounded logarithmic overlap and the box-model estimate (43) $\mathbb{E}_* \log h_n(|Z|^2) = 2\log n + O(1)$ prove the first assertion.

For the relative comparison the output factors cancel exactly. It remains to control $\mathbb{E}\log\|C_p\|_2 - \mathbb{E}\log\|\mathbf{a}\|_2$; a statistic depending only on the polynomial has the same expectation under $\mathbb{P}_*$, because there are always n roots. With $b = (a_0, \dots, a_{n-1})^T/a_n$, the last column gives the lower bound below, while $C_pC_p^* = \operatorname{diag}(0, 1, \dots, 1) + bb^*$ gives the upper bound:

$$\|b\|_2 \le \|C_p\|_2 \le \sqrt{1 + \|b\|_2^2}.$$

Since sums of m independent squared standard complex Gaussians have the $\mathrm{Gamma}(m, 1)$ law, their logarithmic means give

$$\frac{\gamma}{2} - \frac{1}{2n} \le \mathbb{E}\log\|C_p\|_2 - \mathbb{E}\log\|\mathbf{a}\|_2 \le \frac{\gamma}{2}. \tag{58}$$

For example, this follows from $\mathbb{E}\log\mathrm{Gamma}(m, 1) = H_{m-1} - \gamma$, with $H_0 = 0$. Adding (58) proves the relative assertion. □

These are signed logarithmic comparisons of two specified local condition numbers. They say that the geometric mean of their ratio is bounded above and below by constant multiples of n. The comparison does not include errors in forming C_p or in recovering information from an eigenvector; an end-to-end stability theorem would also have to analyze those maps.

6 Interpretation and further questions

The local and companion results describe two different sources of sensitivity. The normalized Jacobian law is determined by a first-order Gaussian calculation and the determinant weight at a root. Its inverse moments have threshold four. The companion overlap additionally contains the coefficient norm of a Lagrange polynomial. For a fixed degree $n \ge 2$ its moment threshold is again four, but degree uniformity has threshold two. The radial estimates and the exact second moment show why concentration of roots near the unit circle does not permit discarding the complementary radii from conditioning averages.

This distinction also explains the relation to the limiting spectral measures in [5, 4]. At a simple eigenvalue λ of a matrix M, the spectral projector satisfies

$$\Pi_\lambda = \lim_{z\to\lambda}(z - \lambda)(zI - M)^{-1} = \frac{rw^*}{w^*r}, \qquad \|\Pi_\lambda\|_2 = \frac{\|r\|_2\|w\|_2}{|w^*r|}.$$

Consequently an overlap estimate requires control of a resolvent residue. Barbarino–Noferini's estimates for shifted monic companions [5, Lemma A.11] concern fixed spectral parameters away from the unit circle; they cannot simply be evaluated at a random eigenvalue, where the shifted matrix is singular. Their decomposition into a deterministic block shift and a Gaussian correction, followed by the Woodbury identity, nevertheless suggests a useful route for studying such residues in block companion models.

6.1 A finite-degree illustration

Table 1 evaluates the exact integral (56). The conditional second moment on the unit circle stays below two, while the unconditional second moment grows. These values illustrate the proved formulas; they are not used to infer either asymptotic or a uniform bound.

The calculation is reproducible from `research/paper-ac/second_moment_table.py`. A separate symbolic script checks the cofactor polynomial, reversal, the conditional variance, and the closed second-moment integrand in small degrees. These computations are consistency checks of the identities, rather than replacements for the proofs.

n	$\Phi_n(1)$	$\mathbb{E}_*\Lambda^2$	$\mathbb{E}_*\Lambda^2 - 2\log n$
1	1.000000	1.000000	1.000000
2	1.333333	1.472800	0.086505
8	1.777778	2.849863	-1.309020
32	1.939394	4.876708	-2.054764
128	1.984496	7.342059	-2.362002

Table 1: Deterministic quadrature of the exact second-moment formula. Positive-coefficient Horner evaluation avoids cancellation at $r = 1$. Computations at 30 and 50 decimal digits agreed to more than 25 decimal places; the table reports six. Degrees one and two also agree with their exact values in Example 5.6.

6.2 What a conditioning comparison establishes

Corollary 5.7 says that the geometric mean of the relative companion-to-coefficient condition ratio is comparable to n. The ratio uses the same scalar output but two different input spaces. It is compatible with bounded average logarithmic eigenvector overlap because the Euclidean coefficient-to-root condition has its own degree scale. Neither statement alone bounds floating-point error in a complete algorithm.

An analysis of a particular implementation would have to control formation of the companion matrix, the structure of the backward error from the eigensolver, and any subsequent recovery. Equation (18) explains the role of structure: a perturbation induced exactly by polynomial coefficients has the original root derivative, whereas an arbitrary matrix perturbation need not arise from a comparably small coefficient perturbation. Other linearizations can change this comparison [26, 14].

The selected-root tail also gives, by a union bound,

$$\mathbb{P}\{\max_{p(z)=0} \Lambda(p, z) > t\} \le \min\{1, Cn/t^2\}.$$

The factor n is part of this bound. Controlling a typical root, an expected fraction of roots, and the worst root are different probabilistic objectives.

6.3 The multivariate problem left by the local theory

For systems, the representation question is now isolated in $\Theta^{\mathcal{B}}_{n,d}$ from (46). The local law and the box metric estimates already hold in every dimension. What remains is to control interpolation angles for an explicitly specified quotient basis or algebraic construction. The univariate division map D_z has a particularly simple tridiagonal Gram matrix; a multivariate replacement must account for relations between polynomial multiples and for basis selection.

Two extensions require additional ideas. First, bounds for a fixed bivariate construction would begin to connect the local theory with multivariate algebraic rootfinding. Any dependence on the dimension must be retained if the objective is to compare with dimension-exponential worst cases. Second, an additive perturbation model $P = P_0 + \sigma Q$, where P_0 is fixed and Q is a centered Gaussian system, produces noncentral conditional derivatives and a noncentral determinant weight. The centered independence law of Theorem 3.3 cannot be transferred to that smoothed model without further analysis. The results proved here provide the centered benchmark for these questions. Smoothed local conditioning and homotopy complexity are already studied in the homogeneous Bombieri–Weyl setting [10, 2]; see also Section 3.4. The remaining task here is to control the specified coefficient and representation conditions for the box ensemble and its noncentered perturbations.

The noncentral Gaussian estimate of [5, Theorem A.5] is a possible ingredient in this task, not a substitute for the determinant-weighted law at a selected root. Moreover, matrices produced by multivariate elimination generally reuse the same input coefficients in several entries. Applying random matrix-polynomial results to them requires treating these dependencies explicitly.

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